

DONALD B. HAVERY

Applied AI Engineer · Production AI, End-to-End · Owner-Operator's Judgment · Agentic Systems · RAG & Context Engineering · Evals · MCP · Local-First

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SUMMARY

Applied AI engineer. I build production agentic systems end-to-end — assistant, tools, evals, fine-tuning — and keep them running. Recent work: **Companion** (local-first voice assistant; 7-provider model router across Claude / Gemini / Qwen / local Ollama; 12-persona system), **Albums** (semantic search + RAG over ~3,600 indexed items), and **Wire** (an unattended pipeline that researched and published a daily LinkedIn brief — 35 posts in production). 38 repos, 3,300+ commits, 5,000+ tests authored; heavy Claude Code user with 25+ skills of my own. I've also founded and run my own businesses end-to-end and led inside federally regulated, safety-critical operations — including certifying new operators to a federal standard — so I build software already knowing what it has to do for the business that depends on it.

REPRESENTATIVE PROJECTS

A local-first voice assistant with a 7-provider model router (Companion)

Python · Rust + Tauri · PySide6 · Android · 695 commits · 345 tests · dbhavery.dev/companion

- **7-provider model router** (Claude, Gemini, Qwen, local Ollama, OpenAI, Groq, OpenRouter) with 3-tier complexity routing — regex, then keyword, then an LLM classifier — choosing the right model per request
- **Context engineering:** ChromaDB hybrid retrieval (BM25 + dense embeddings), tiered memory, MCP integration, async tool execution
- **12-persona system** with consistent persona, safety, and tool-use behavior across providers, and failover between them
- Local-first by design with cloud failover; sub-500 ms latency budget; circuit breakers, bulkhead isolation, and retry budgets

A behavioral eval harness that catches an agent's regressions before they ship (Quality Eval Harness)

Python · 18 eval cases · 6 quality domains · CLI-driven

- Scores the agent against an 18-case suite spanning 6 quality domains, with adversarial and canary cases that catch regressions before they ship
- **Grader pass** audits whether the eval assertions themselves discriminate — catches tests that pass for the wrong reasons. Meta-evaluation as a first-class component, not an afterthought
- Tracks failure modes like “refusal theatre” and tool-use drift, so quality changes surface in CI rather than in production

A self-hosted AI operations platform — six independent packages (Citadel)

Python · FastAPI · HNSW (pure NumPy) · 100 tests · MIT · live demo on Hugging Face

- Replaces managed LLM infrastructure with six independent pip packages: a multi-provider gateway (Claude / Gemini / OpenAI / Ollama) with **circuit-breaker failover**, a from-scratch **HNSW vector engine** in pure NumPy (no FAISS / pgvector), a ReAct agent runtime, document ingestion, and cost/latency observability
- Built the vector index from the Malkov & Yashunin (2018) paper — $O(\log n)$ approximate nearest-neighbor, zero external vector-DB dependency, \$0/month vs \$25–70 managed

A self-hosted semantic file library with RAG chat (Albums)

FastAPI · React · ChromaDB · sentence-transformer embeddings

- Self-hosted Google Photos / Drive replacement: semantic search and RAG chat over ~3,600 indexed media items — the same retrieval patterns Companion uses, on a larger personal corpus

An autonomous research-to-publication pipeline (Wire)

Python · ran in production (35 posts), now parked — proof complete

- Self-reviewing research pipeline that pulled from ~18 sources (17 feeds plus a rotating web-search layer), drafted, fact-checked, generated a hero image, and auto-published to LinkedIn twice a day with no human in the loop — 35 posts before I parked it, having proved the unattended loop. Reliability discipline on a daily cron

Reusable Claude Code skills & agent specs across projects

25+ authored skills and custom agent definitions

- Reusable Claude Code skills and agent specs used across my projects. Each spec sets explicit refusal boundaries, escalation paths, and verification gates, so the rare unsafe case is caught before it ships

TECHNICAL SKILLS

Agentic Systems	Multi-agent orchestration , sub-agent coordination, intent routing, autonomous loops, long-horizon agent supervision, tool use / function calling at production scale, behavioral specifications, fallback chains
RAG & Retrieval	Hybrid retrieval (BM25 + dense), ChromaDB, sentence-transformer embeddings, tiered memory, semantic search, context engineering, retrieval pipelines
Evals & Observability	Deterministic offline assertions (forbids / requires / length / tone), adversarial & canary probes, meta-evaluation (the probes test the tests), regression detection across model snapshots, CI exit-code gating, observability for AI
MCP & Tool Use	Model Context Protocol (production integrations), tool harnesses, function calling at scale, async tool execution, custom MCP servers
Claude Code	Heavy Claude Code user — 3,300+ commits authored through Claude Code, 25+ skills and custom agents of my own, plugin development
LLM Stack	Claude (Opus/Sonnet/Haiku), Gemini (Pro/Flash), Qwen, Ollama local serving, cross-provider behavioral parity, latency budgets, circuit breakers
Reliability	Circuit breakers (Closed/Open/Half-Open), bulkhead isolation, retry budgets, sub-500 ms voice latency budget, observability for AI
Backend	Python (asyncio, FastAPI), Rust (Tauri, tokio, serde), Node.js, WebSocket, PostgreSQL, SQLite, Redis, Docker, GitHub Actions
Frontend / Product	Next.js, React 19, TypeScript, Vite, Tailwind, Three.js / R3F, GSAP, PySide6 desktop
Fine-tuning	LoRA / QLoRA training pipelines (image-domain LoRA shipped), HuggingFace, PyTorch
Operations & Ventures	Founder / owner-operator (multiple solo ventures): business operations & planning, DTC e-commerce, product formulation & small-batch manufacturing, brand & packaging design, digital marketing, supply chain, fulfillment & logistics, inventory, customer service; investigative research, surveillance & case management; HAZMAT-certified regulated transport (DOT/FMCSA), risk & safety management

EXPERIENCE

AI Engineer — Independent

Current

- Design and ship production agentic systems end-to-end: harness, tools, evaluation, fine-tuning. 38 repos, 3,300+ commits, 5,000+ automated tests authored
- Built a calibrated eval harness (18 cases across 6 quality domains, with adversarial and canary probes) that catches quality and tool-use regressions in CI, before they reach production
- Operate as a Claude Code pair-programmer daily; keep prompts, evals, and harnesses in version control and under test, the same as any other production code
- Public portfolio: dbhavery.dev (Next.js + R3F + GSAP, project showcases with real artifacts)

Founder & Owner-Operator — Independent Ventures

- **Original Lumberjack** — premium men's grooming, DTC e-commerce. Ran the whole business solo: formulated products from raw ingredients, manufactured in small batches, designed the brand, packaging, and labeling, and handled marketing, fulfillment, shipping, and customer service
- **Precision Investigations** — independent case work: investigative research, surveillance, case management, evidence handling, client reporting, and strict confidentiality
- **Sta-Full** (fleet-fueling logistics) and **angrynirds** (a remote, contract AI-engineering collective, currently building) — scoped, branded, and stood up independently
- Every venture built and run end-to-end by one person, alongside full-time work — the same ship-it-yourself instinct I bring to production AI

Operations & Safety-Critical Logistics — Certified Trainer

Federally regulated, high-consequence operations under DOT/FMCSA oversight — HAZMAT-certified, zero-error standards, real liability. Certified to train and sign off new operators to a federal standard: documented process, repeatable judgment, and clear accountability when something goes wrong. The same discipline — process you can document, decisions you can defend, verification before anything ships — carries directly into how I build and run production AI.

EDUCATION

Capella University — B.S. Computer Science (in progress).

CERTIFICATIONS

Anthropic Academy — 15 courses complete (Building with Claude, Claude Code, Claude Code Skills, MCP & MCP Advanced, Cowork, Bedrock, Vertex AI, AI Fluency suite, AI Capabilities & Limitations)

DeepLearning.AI — Generative AI with Large Language Models | **Google** — AI Fundamentals (Coursera-verifiable)

Sophia Learning — 30 ACE college-credit courses, verified (public Credly profile) — CS foundations (Python, Java, web dev, SQL/databases, networking, statistics) through business and analytics